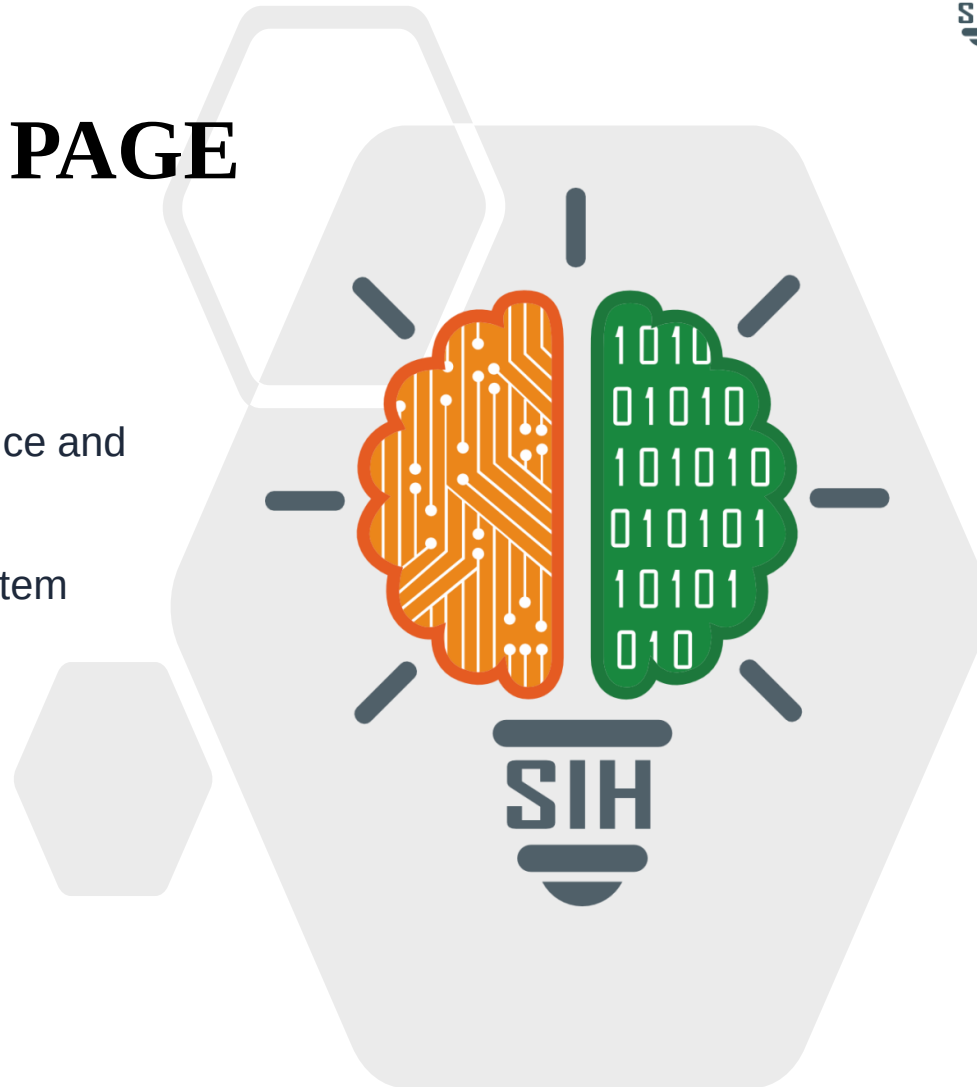


SMART INDIA HACKATHON 2026

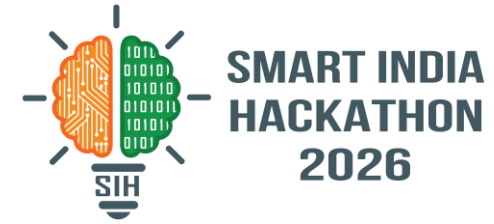


TITLE PAGE

- **Problem Statement ID** – SIH26024
- **Problem Statement Title** – AI-Based Smart Governance and Compliance Monitoring System for Coal Mines
- **Theme** – Mining and Mineral Transport Monitoring System
- **PS Category** – Software
- **Team ID** – SIH-S-B3-147
- **Team Name (Registered on portal)** – Codeavengerz



IDEA TITLE: SMARTMINEGUARD



❖ Proposed Solution (Describe your Idea/Solution/Prototype)

• Detailed explanation of the proposed solution:

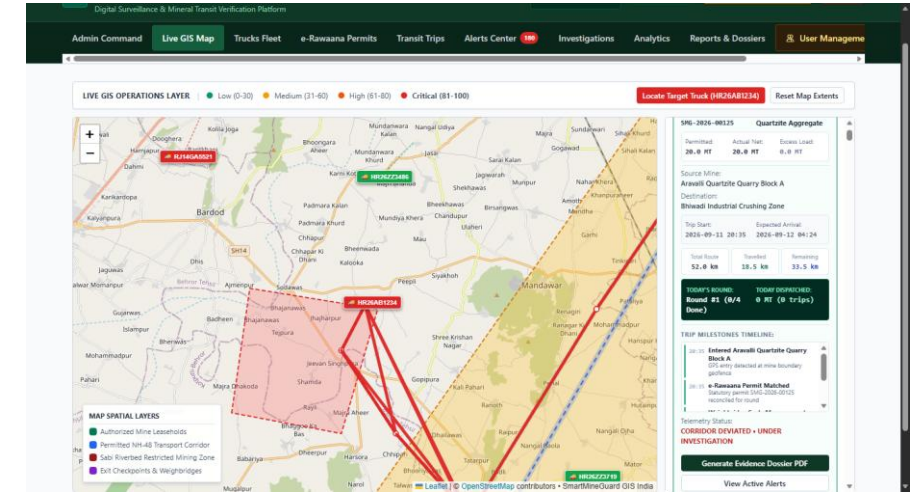
- Pure-software statutory intelligence & governance ecosystem for State Mining Depts, DGMS & Coal Controllers.
- Unifies e-Rawaana transit passes, AIS-140 GPS telematics, digital weighbridges & mine lease quotas into one digital audit trail.
- Working production-grade prototype with role-based portals (Admin, Field Officer, Mine Operator).

• How it addresses the problem:

- Eliminates Illegal Extraction: Reconciles mine lease extraction quotas against cumulative dispatch tonnage in real-time.
- Stops Transit Pass Fraud: Single-use encrypted QR tokens mark passes CONSUMED at checkpoints, preventing reuse.
- Halts Mineral Theft & Deviation: Live AIS-140 GPS tracking detects corridor departures, unapproved halts & geofence breaches.
- Enforces Axle Weight Limits: Automated weighbridge integration flags gross vs. tare weight variances (>5%) before dispatch.

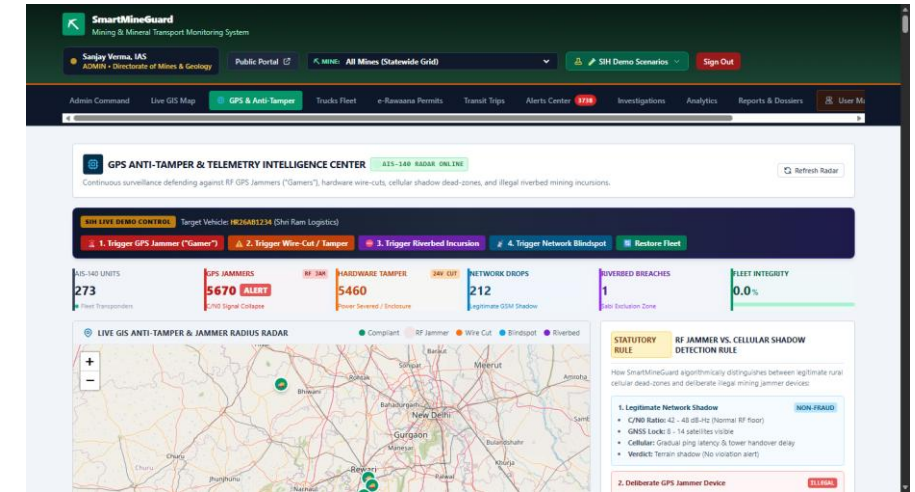
• Innovation and uniqueness of the solution:

- 100% Deterministic & Defensible: Explainable 6-rule scoring (0-100) compliant with MMDR Act; zero AI hallucinations.
- Instant Field Enforcement PWA: Mobile QR scanner reconciles permits in <2s with offline validation capability.
- Court-Ready Evidence Dossiers: Auto-generates cryptographically verifiable, timestamped PDF dossiers for legal prosecution.



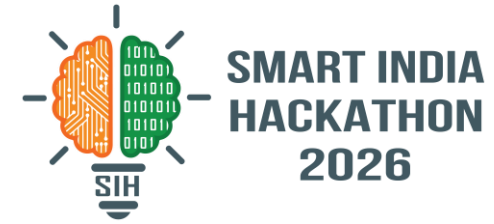
Live GIS Fleet Surveillance & Corridor Deviation
[Working Prototype: PostGIS Spatial Corridors & Real-Time Tracking]

- Technologies to be used (programming languages, frameworks, hardware):**
 - Core Backend: Python 3.13, Flask 3.0, Flask-SocketIO (bi-directional WebSocket real-time fleet telemetry).
 - Spatial Database: PostgreSQL 16 + PostGIS for spatial vector analytics (ST_Contains, ST_DWithin, corridor buffers) + zero-setup SQLite spatial fallback.
 - Frontend Architecture: HTML5, Jinja2, Vanilla JavaScript (ES6), Tailwind CSS, Leaflet.js 1.9 (GIS layers), Chart.js 4.4.
 - Enforcement & Security: ReportLab 5.0 (Court-ready PDF dossiers), HTML5-QRCode, Cryptographic SHA-256 tokens.
 - Hardware & IoT: AIS-140 GPS telematics protocol, RS-232/IP Digital Weighbridge Scales, ANPR cameras, RFID boom barriers.
- Methodology and process for implementation (Flow Charts/working prototype):**
 - End-to-End Operational Pipeline:
 - [Lease Quota Check] → [e-Rawaana Pass (QR)] → [IoT Weighbridge & ANPR] → [Live AIS-140 GPS Tracking] → [6-Rule Detection Engine] → [Explainable Risk Score (0-100)] → [Roadside QR Check] → [Court Evidence Dossier PDF].
 - Deterministic Detection Engine: Evaluates 6 statutory rules (Route Deviation, Speed Anomaly, Lease Geofence Breach, Weighment Variance >5%, GPS Blackout, Expired Permit).
 - Working Prototype: Fully implemented with 26+ screens, multi-truck GPS simulator, and court-ready PDF dossier engine.



AIS-140 GPS Telemetry & Live Telematics Terminal
[Working Prototype: SocketIO Vehicle Movement & Sensor Stream]

FEASIBILITY AND VIABILITY



• Analysis of the feasibility of the idea:

- Technical Feasibility: Proven open-source stack (Python/PostGIS) with minimal compute footprint; runs on standard NIC/cloud servers without GPU costs.
- Operational Feasibility: REST APIs integrate seamlessly with Khanij Online, e-Challan, Parivahan Vahan/Sarathi, and existing physical weighbridges.
- Financial Viability: Zero proprietary licensing or recurring AI token fees; rapid ROI achieved by plugging royalty revenue leakage.

• Potential challenges and risks:

- Network Blindspots: Remote mining valleys and forest transit corridors experience cellular signal loss.
- Hardware Tampering: Deliberate unplugging of AIS-140 GPS trackers or weighbridge scale manipulation.
- Enforcement Resistance: Operator reluctance and attempts to pass forged physical challans.

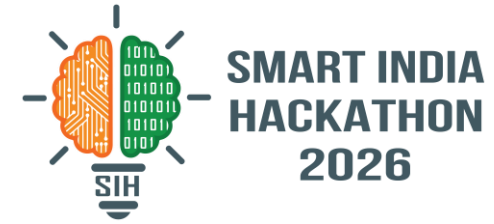
• Strategies for overcoming these challenges:

- Store-and-Forward Telemetry: Onboard device buffers pings; offline PWA cryptographic QR verification for field officers without internet.
- Multi-Sensor Cross-Validation: Anomaly engine cross-references GPS ping continuity, velocity spikes, and scale weight variance (>5%).
- Cryptographic Lifecycle: Passes auto-expire with strict single-use states (ISSUED → IN_TRANSIT → CONSUMED).

Alert ID	Logged At	Vehicle / Permit	Detection Type	Rule-Based Score	Severity	Investigation Status	Assigned Officer	Statutory Enforcement Actions
ALT-20260912022354-EXC	2026-09-11 20:53:54	R214GAS521 TIP-2026-08324	Excessive Trip Frequency EXCESSIVE TRIP FREQUENCY: Truck R214GAS521 completed 27 rounds against an allowed limit of 4 (23 unpermitted extra rounds).	+25	HIGH	NEW	Inspector Rajesh K. Meena	Acknowledge Elevate to Case Dismiss
ALT-20260912022353-EXC	2026-09-11 20:53:53	00020AB012 TIP-2026-08880	Excessive Trip Frequency EXCESSIVE TRIP FREQUENCY: Truck 00020AB012 completed 25 rounds against an allowed limit of 4 (21 unpermitted extra rounds).	+25	HIGH	NEW	Inspector Rajesh K. Meena	Acknowledge Elevate to Case Dismiss
ALT-			Unauthorized Mine Entry UNAUTHORIZED MINE ENTRY: ...					Acknowledge Elevate to Case

Real-Time Alert Center & Statutory Investigation
[Working Prototype: Automated Risk Categorization & Elevation]

IMPACT AND BENEFITS

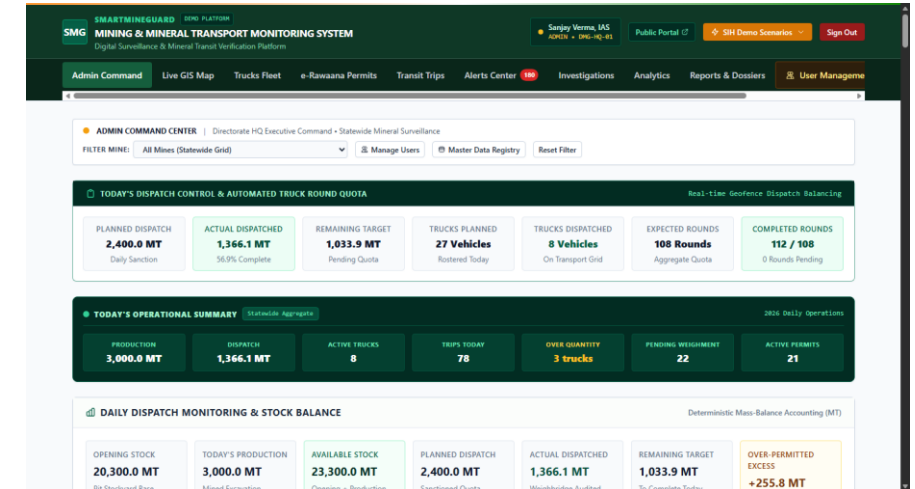


- **Potential impact on the target audience:**

- State Mining Departments & Coal Controller: 100% transparent auditability of extraction vs. dispatches; stops ₹100s of Crores in royalty leakage.
- Field Enforcement Officers: Instant roadside verification in <2 seconds; automated evidence dossiers save hours of manual documentation.
- Mine Leaseholders & Transporters: Automated weighbridge clearance (<30 seconds) with automated barrier lift, eliminating transit bottlenecks.

- **Benefits of the solution (social, economic, environmental, etc.):**

- Economic Impact: 100% royalty recovery, elimination of transit pass recycling, and eradication of unrecorded commercial dispatches.
- Environmental Protection: Strict lease boundary geofencing halts illegal excavation in riverbeds, forest reserves & eco-sensitive zones.
- Public Safety & Infrastructure: Enforces strict Gross Vehicle Weight limits to prevent road damage and fatal overloading accidents.
- Governance & Accountability: Tamper-evident digital audit trail eliminates bribery and ensures compliance under the MMDR Act.



Executive Command & State Governance Dashboard
[Key Metrics: 100% Auditability | <2s Field Check | Zero Pass Reuse]

- Details / Links of the reference and research work:**

- Statutory Frameworks & Regulatory Guidelines:

- Mines and Minerals (Development and Regulation) Act (MMDR Act, 1957 & 2023 Amendment) – Statutory transit rules.

- Ministry of Coal & Ministry of Mines – Guidelines for Sustainable Mining & Dispatch Surveillance.

- Central Motor Vehicles Rules (CMVR) Rule 125H – Mandating AIS-140 Vehicle Location Tracking (VLTD).

- Technical Standards & Specifications:

- AIS-140 Standard for Intelligent Transportation Systems & Telematics Protocols (ARAI / ICAT).

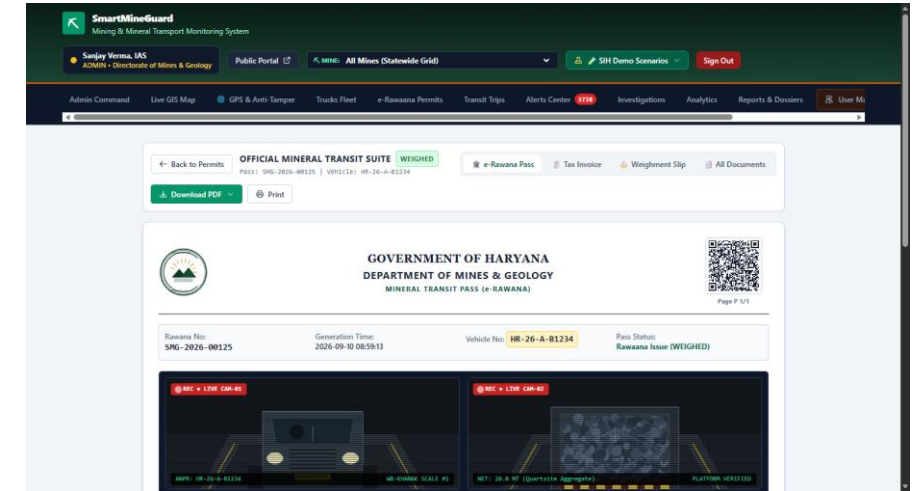
- Open Geospatial Consortium (OGC) Simple Features for SQL – PostGIS spatial indexing (ST_Contains, ST_DWithin).

- Legal Metrology (Enforcement) Rules – Electronic weighbridge integration & tare accuracy standards.

- Working Prototype Repository & Verification:

- Complete Functional Prototype: Python/Flask, PostGIS, Leaflet GIS, SocketIO, ReportLab Dossier Engine.

- Tested on Real Mining Corridors: 26+ screens, 6 compliance algorithms, real-time telemetry & court-ready dossiers.



Official e-Rawaana Pass & Legal Dossier Reconciliation

[Statutory Audit Trail: SHA-256 Token & Legal Metrology Verification]